CONTEXTUAL DECISION MODELLING AND ADAPTIVE AI SYSTEMS

25-26J-145-DS

Project Proposal Report

Shahaam M M

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date

31.08.2025

Mr. Samadhi Rathnayake

ABSTRACT

This project develops a context-aware, multimodal decision-prediction system for neuromarketing that adapts in real time to changing user and environmental states. We introduce a context layer that conditions model behavior on situational variables (e.g., product category, device, time), environmental proxies, and coarsened demographics, alongside behavioral signals and optional affective/physiological inputs. The layer is integrated with a multimodal backbone (text, image, audio, interaction streams) and adaptive learning mechanisms—meta-learned user adapters for fast personalization and a contextual-bandit policy that improves content or action selection from continuous feedback. Methodologically, we combine self-supervised pretraining, context-conditioned modulation (FiLM/hypernetwork adapters), and online updates with drift detection and safety guardrails. We will evaluate across varied consumer contexts, cold starts, and missing/noisy modalities using accuracy (AUROC/AUPRC, log loss), calibration (Brier score, ECE), cumulative regret, robustness to distribution shift, and fairness across demographic bands. Anticipated outcomes include higher predictive accuracy and better calibration than context-agnostic baselines, faster personalization with reduced regret, and stable performance in the presence of noisy or absent modalities. The work bridges personalization with neural–behavioural data fusion and demonstrates practical use of reinforcement and meta-learning to move neuromarketing models from lab settings toward reliable, privacy-aware, real-world deployment.

Keywords: context-aware modelling; multimodal fusion; reinforcement learning; meta-learning; calibration.

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1. INTRODUCTION

In recent years, advances in artificial intelligence (AI) and neuromarketing have transformed the way consumer decision-making is modeled and predicted. Traditional approaches, however, largely rely on static personalization techniques that emphasize user profiles or isolated behavioral signals. While these models provide valuable insights, they fail to account for the dynamic nature of decision-making, which is often shaped by contextual, emotional, and environmental variables. As a result, existing systems risk oversimplifying consumer behavior and limiting their predictive power in real-world applications.

Emerging research in multimodal AI and affective computing demonstrates the value of integrating physiological, emotional, and behavioral signals to gain richer consumer insights [1]. Similarly, context-aware frameworks are beginning to reveal how situational cues can modulate consumer responses, highlighting the importance of embedding environmental and demographic factors into decision models [2]. Yet, despite these advances, neuromarketing systems remain constrained by lab-based conditions, limited adaptability, and insufficient integration of continuous feedback loops.

This research proposes a contextual decision modeling framework that introduces an adaptive, context-aware layer into neuromarketing AI systems. By fusing neuro-behavioral data with environmental and demographic variables, and embedding reinforcement and meta-learning mechanisms, the system dynamically adjusts decision predictions in real time. This design moves beyond static personalization toward human-like adaptability, enabling models that evolve continuously as consumer contexts shift. In doing so, the proposed framework bridges the gap between controlled neuromarketing studies and deployable real-world applications, offering a robust foundation for next-generation consumer AI.

1.1. Background and Literature Survey

1.1.1 Neuromarketing and AI-Driven consumer prediction

Neuromarketing combines insights from neuroscience and marketing to understand how consumers respond to products, brands, and advertisements. Early neuromarketing approaches primarily relied on neurophysiological measures such as EEG, eye-tracking, and galvanic skin response (GSR) to infer emotional engagement and attention. While effective in controlled experiments, these methods have historically lacked scalability for real-world environments.

Recent advances in AI-enhanced neuromarketing have demonstrated the power of multimodal data integration. For example, César et al. [1] show how affective computing and multimodal AI can combine cognitive, emotional, and behavioral signals to improve consumer insight prediction. Similarly, Overgoor et al. [3] propose a NeuroAI framework where contextual embeddings explain the influence of visual stimuli on consumer decision-making. These works highlight a shift toward richer data fusion, but remain largely static and lab-based.

Counterfactual explanations have gained traction as a way to generate actionable insights by exploring “what-if” scenarios [7]. In marketing applications, counterfactual reasoning

1.1.2 Context-Aware AI in consumer decision modelling

Context-aware AI refers to systems that adjust their predictions based on situational variables such as time, environment, location, or emotional state. In domains like recommendation systems, contextual bandits and context-aware embeddings have been shown to improve personalization and user engagement. [2] demonstrate how context-aware AI enhances decision intelligence by dynamically integrating environmental data into prediction models. Likewise, in e-commerce,[4] illustrate how context-aware recommendation engines improve the quality of personalized offers by refining predictions with real-time contextual cues. Despite these advances, neuromarketing applications rarely embed contextual adaptation. Most models focus on static personalization without incorporating the dynamic situational context of consumer decision-making.

1.1.3 Adaptive learning in consumer AI systems

Adaptive AI leverages reinforcement learning (RL) and meta-learning to evolve continuously as users interact. Reinforcement learning has been widely used in domains such as recommendation systems and trading decision models, enabling systems to optimize long-term engagement and adapt to user preferences in real time. Meta-learning approaches such as Model-Agnostic Meta-Learning (MAML) enable systems to generalize to new users or contexts with minimal retraining. Reviews of AI in consumer behavior identify reinforcement and meta-learning as critical directions for building more adaptive, human-like decision models [5][6]. However, most implementations of RL and meta-learning in consumer systems are limited to behavioral data streams. Their integration with neuromarketing signals—particularly physiological and affective data—remains largely unexplored.

1.1.4 Emerging use of generative and NeuroAI in marketing

Generative AI enables dynamic, context-aware advertising tailored to individual users based on emotional and situational cues [7]. Similarly, Pradeep et al. [8] propose a NeuroAI framework that uses attention-based architectures to fuse neuroscience data with generative modeling for consumer targeting. These innovations demonstrate the potential of AI to create deeply personalized, emotionally resonant consumer experiences, yet they often lack adaptability: models generate personalized content but do not evolve continuously in response to shifting consumer states.

1.2. Research Gap

Recent advances in neuromarketing and AI have expanded the ability to analyze consumer behavior, several critical gaps remain. First, existing models are predominantly static in nature, relying on fixed user profiles or isolated neural-behavioral signals. Such approaches overlook the dynamic influence of situational and contextual factors—including environment, demographics, and emotional state—that shape consumer decision-making in real time.

Second, there is limited integration across multimodal data streams. Current systems often treat neural measures (e.g., EEG, GSR), behavioral cues (e.g., eye-tracking, dwell time), and demographic or environmental information in isolation. This fragmentation restricts the development of holistic, human-like decision models capable of capturing the complex interactions between cognitive, affective, and situational variables.

Third, while adaptive AI approaches such as reinforcement learning and meta-learning have shown promise in fields like recommender systems and trading, their application in neuromarketing remains underdeveloped. Most consumer prediction models lack the ability to continuously evolve through user interaction and feedback, resulting in limited adaptability and poor generalization across contexts.

Finally, neuromarketing research continues to be constrained by laboratory-based settings. This reliance on controlled conditions hinders the scalability and real-world applicability of models, which are often unable to perform reliably in natural consumer environments. Ethical and privacy considerations around the collection of sensitive affective and neural data further complicate real-world deployment.

Taken together, these gaps reveal the absence of a context-aware, adaptive neuromarketing framework that integrates multimodal consumer signals with environmental and demographic variables, and evolves dynamically through continuous feedback. Addressing this gap offers the potential to advance both the theoretical understanding of consumer behavior and the practical development of deployable, human-like AI decision systems for real-world marketing contexts.

1.3. Research Problem

Consumer decision-making is inherently dynamic, contextual, and multidimensional, influenced not only by neural and behavioral processes but also by situational, environmental, and demographic factors. Current neuromarketing models, however, remain limited in three major ways. They rely on static personalization approaches, which fail to account for rapid changes in consumer state and context. They treat neural, behavioral, and contextual data streams in isolation, restricting the ability to model consumer decisions holistically. They lack adaptive learning mechanisms, preventing real-time evolution of predictions as consumers interact with changing environments. As a result, existing systems struggle to capture the fluid and context-dependent nature of consumer decision-making. This not only reduces predictive accuracy but also restricts the scalability of neuromarketing applications beyond controlled laboratory conditions.

The research problem therefore centers on the absence of a context-aware, adaptive AI framework that can dynamically integrate multimodal signals with situational variables and evolve continuously through feedback. Without such a framework, neuromarketing remains constrained to static and limited models, unable to fully support real-world, personalized consumer engagement..

2. OBJECTIVES

2.1. Main Objective

Design, build, and validate a context-aware, adaptive multimodal decision-prediction and action-selection framework for neuromarketing that conditions predictions on situational, environmental, and demographic variables; learns online from continuous user feedback, and maintains accuracy, calibration, fairness, and robustness in real-world-like settings.

2.2. Specific Objectives

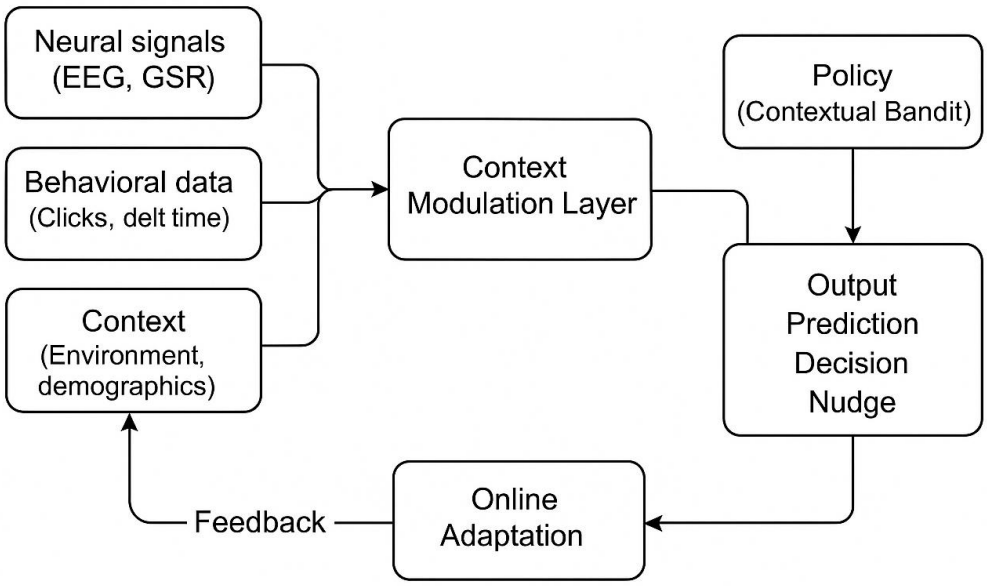
1. Develop a privacy-preserving context encoder that captures product category, device, time, ambient/environmental proxies, and coarsened demographics; produce a compact context vector usable at inference
2. Implement text/image/audio/behavioural encoders with cross-attention fusion that is resilient to missing/noisy modalities.
3. Introduce meta-learned user adapters (e.g., LoRA/BitFit initialised via MAML-style training) for few-shot adaptation.
4. Build a contextual-bandit layer to select creatives/sequences/nudges and learn from rewards (clicks, conversions, ratings).

3. METHODOLOGY

3.1. System Overview

The system adopts a multimodal, context-aware deep learning approach for real-time, personalized consumer decision prediction and action selection. Behavioral signals, optional affect/physiology, content features, and situational context are fused inside a transformer-based backbone with a contextual modulation layer. An adaptive learning stack—meta-learning plus contextual bandits—lets the model evolve continuously from feedback.

3.2. System Diagram



3.3. Tools and Technologies

Category	Tools
Data Gathering and Preprocessing	EEG Datasets, Pandas, NumPy
Modeling	Scikit-learn, TensorFlow, PyTorch
Explainability Methods	SHAP, LIME, Integrated Gradients, Captum (PyTorch)
Counterfactuals	Dice (Python), Custom scenario modeling
Visualization	Python (Plotly, Matplotlib), Power BI, Tableau
Ethical Assessment	Fairlearn, AIF360 (bias and fairness libraries)

3.4. Data Requirements

- Neural Data – EEG datasets that include emotional and cognitive responses.
- Behavioral Data – Decision-making records such as consumer choices, reactions, ratings, or preferences.
- Integrated Data - Behavioral labels mapped to EEG time windows to create multimodal features.

Data will be drawn from public freely available neuromarketing datasets and may be complemented with a small-scale survey or experiment for contextual validation like simple product choice tasks.

3.5. Gantt Chart

ID	Task	2025-2026																							
		June		July		August		September		October		November		December		January		February		March		April		May	
1	Topic Selection																								
2	Topic Assesment																								
3	Literature Review																								
4	TAF																								
5	TAF Evaluation																								
6	Charter																								
7	Proposal Draft																								
8	Proposal presentation																								
9	Propsal Report																								
10	System Development																								
11	Progress Presentation 01																								
12	Reserch Paper																								
13	Progress Presentation 02																								
14	Final Report Submission																								
15	Final Report Feedback																								
16	Final Presentation and Viva																								

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

- **Contextual Data Integration :**
  - Capture and process multimodal data including neural-behavioral (e.g., EEG, GSR, emotion recognition), demographic, and environmental inputs.
  - Represent these inputs in a unified feature space through contextual embeddings.
- **Dynamic Decision Prediction:**
  - Generate consumer decision predictions that adapt in real time to situational and personal variables (e.g., product category, emotional state).
  - Provide probabilistic outputs to reflect uncertainty in predictions.
- **Adaptive Learning Mechanisms:**
  - Incorporate reinforcement learning for continuous improvement based on explicit (e.g., purchase, rating) and implicit (e.g., dwell time, hesitation) feedback.
  - Employ meta-learning to quickly adapt to new users and contexts with minimal retraining.
- **Multimodal Fusion:**
  - Implement early or late fusion strategies (e.g., transformers, attention mechanisms) to combine signals from multiple modalities.
  - Synchronize time-series inputs (e.g., neural signals with behavioral actions).
- **Feedback Loops:**
  - Continuously collect and integrate user responses to refine predictions.
  - Enable self-adjustment of weights and parameters for personalization.
- **Evaluation and Monitoring:**
  - Provide performance metrics such as accuracy, adaptability, and robustness.
  - Log system decisions and contextual conditions for traceability and analysis.

4.2. Non-Functional Requirements

- Real-Time Performance: Decision predictions and adaptations should be generated with low latency for seamless integration into real-world applications (e.g., e-commerce, AR/VR).
- Robustness: Must maintain accuracy under noisy, incomplete, or missing data conditions, and should handle variability in consumer responses without significant degradation in performance.
- Usability: Interfaces for marketers/researchers must be intuitive, with clear visualization of decision predictions and contextual insights.
- Ethical and Privacy Compliance Ensure transparent data handling and anonymization of sensitive neuro-behavioral information Provide mechanisms for user consent, data minimization, and compliance with regulations (e.g., GDPR).

4.3. User Requirements

- Researchers: Need in-depth explanations and detailed visualizations for validation.
- Marketers: Require simplified, actionable insights.
- Consumers: Need transparency in how their decisions are being influenced.
- Regulators: Require ethical scoreboards to monitor manipulation risk.

4.4. System Requirements

Hardware	• A computer with at least 16 GB RAM, GPU support (NVIDIA) for deep learning. • Storage capacity of 1 TB.
Software	• Python 3.x • ML/DL frameworks (TensorFlow, PyTorch) • Explainability libraries (SHAP, LIME, Captum, DiCE) • Visualization tools (Python Libraries, Power BI, Tableau) • Fairness libraries (AIF360, Fairlearn)

4.5. Test Cases (Tentative)

1. Contextual Adaptation
    - Test Case: Vary consumer's emotional state (e.g., calm vs. stressed) while presenting the same product.
    - Expected Outcome: The system adjusts decision prediction probabilities to reflect the context (e.g., reduced purchase likelihood under stress).
    - Requirement Validated: Contextual Data Integration & Dynamic Decision Prediction.
2. Multimodal Fusion
    - Test Case: Provide only EEG data vs. EEG + eye-tracking + demographic info.
    - Expected Outcome: The multimodal input yields higher accuracy than single-modality input.
    - Requirement Validated: Multimodal Fusion & Generalizability..
3. Adaptive Learning Over Time
    - Test Case: Simulate repeated interactions with a consumer whose preferences shift gradually (e.g., from eco-friendly products → luxury items).
    - Expected Outcome: The system adapts prediction outputs over sessions, reflecting preference shifts without retraining from scratch.
    - Requirement Validated: Adaptive Learning Mechanisms & Robustness."
4. Cold-Start User (Meta-Learning Test)
    - Test Case: Introduce a new user with minimal data.
    - Expected Outcome: The system generates reasonable initial predictions by leveraging demographic and contextual similarity to prior users.
    - Requirement Validated: Meta-Learning & Scalability.
5. Noisy/Incomplete Data
    - Test Case: Remove or corrupt part of the EEG or behavioral dataset.
    - Expected Outcome: The system maintains stable predictions by relying on available contextual and demographic data.
    - Requirement Validated: Robustness & Real-Time Performance.
6. Real-Time Responsiveness
    - Test Case: Stream live user interactions (e.g., browsing in an online shop).
    - Expected Outcome: Decision predictions update in under 10 milliseconds, supporting seamless real-time integration.
    - Requirement Validated: Real-Time Performance.
7. Ethical Compliance
    - Test Case: Request consumer consent → user declines.
    - Expected Outcome: The system halts data collection and anonymizes any cached data.
    - Requirement Validated: Ethical & Privacy Compliance

5. DESCRIPTION OF PERSONAL AND FACILITIES

5.1. Personal Contribution

I will lead the design, development, and validation of a context-aware, adaptive AI framework for neuromarketing, including multimodal data integration, contextual modulation, meta-learning, and reinforcement learning for dynamic decision prediction. Responsibilities include data collection and preprocessing, model development, performance evaluation, and ethical handling of sensitive data. Computational resources include high-performance GPUs and cloud platforms (AWS, Google Cloud, Azure) for scalable training, alongside Python-based software tools (PyTorch, TensorFlow, HuggingFace Transformers) and visualization libraries (Pandas, Matplotlib, Seaborn). Data resources comprise anonymized consumer behavior datasets and multimodal content, with optional lab-based neurophysiological data. Guidance from faculty supervision and access to collaborative platforms ensure robust project execution.

5.2. Facilities and Resources

- Supervisory Support:
  1. Supervisor - Mr. Samadhi Rathnayake
  2. Co-Supervisor – Ms. Samanthi Siriwardene
  3. External Supervisor – Dr. Kamal Gunarathne
- Software Tools: Open-source libraries including PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, and visualization tools: Pandas, NumPy, Matplotlib, Seaborn and version control such as Git.
- Data Resources: Publicly available EEG and behavioral datasets for consumer decision-making scenarios.
- University Facilities: University-provided resources such as GitHub Enterprise, cloud storage, access to the university's digital library, research databases, and journal subscriptions for literature review.

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